CS 5361/6361 Machine Learning
Spring 2026

Instructor: Olac Fuentes

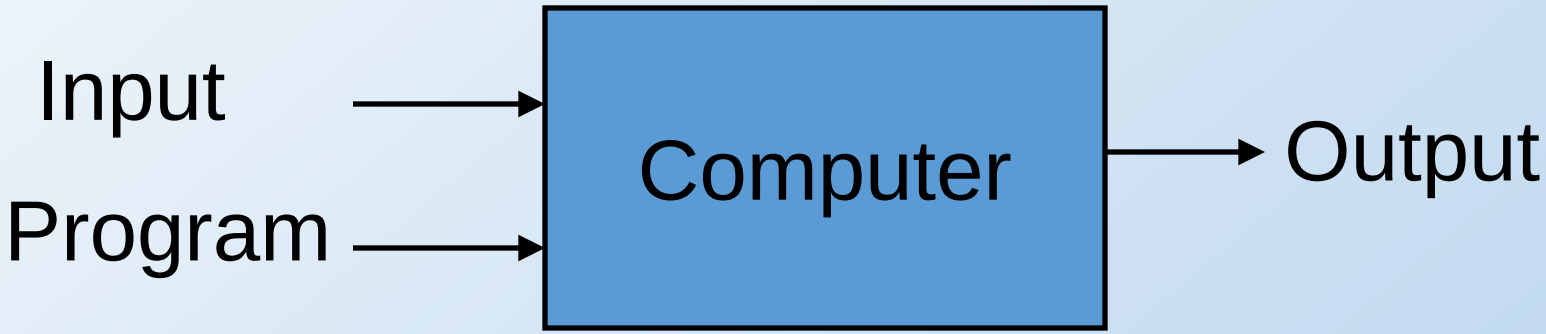
Required Background

- Programming ability at least corresponding to successfully completing CS2302
- Python, particularly working with 2D and 3D arrays
- Probability and Statistics
- Linear algebra

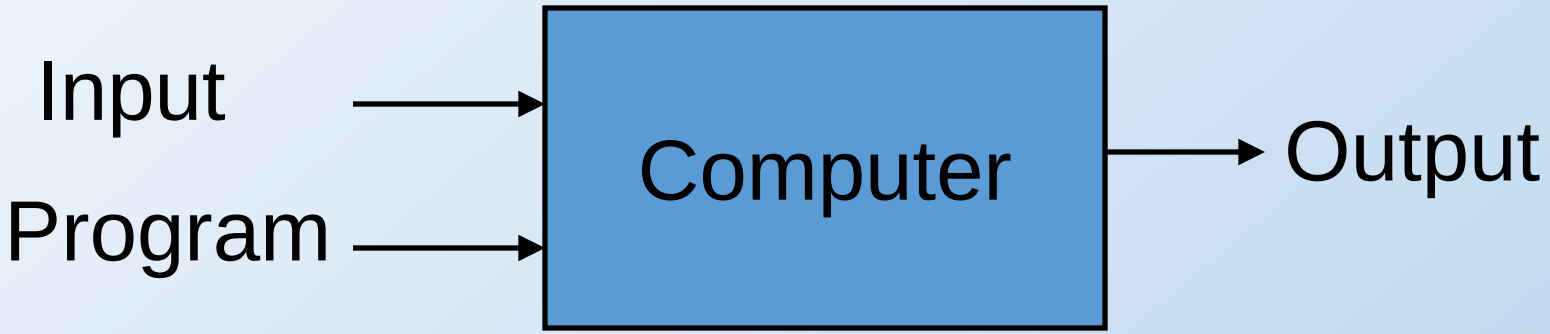
What Is Machine Learning?

- Automating automation
- Getting computers to program themselves
- Writing programs that improve with experience

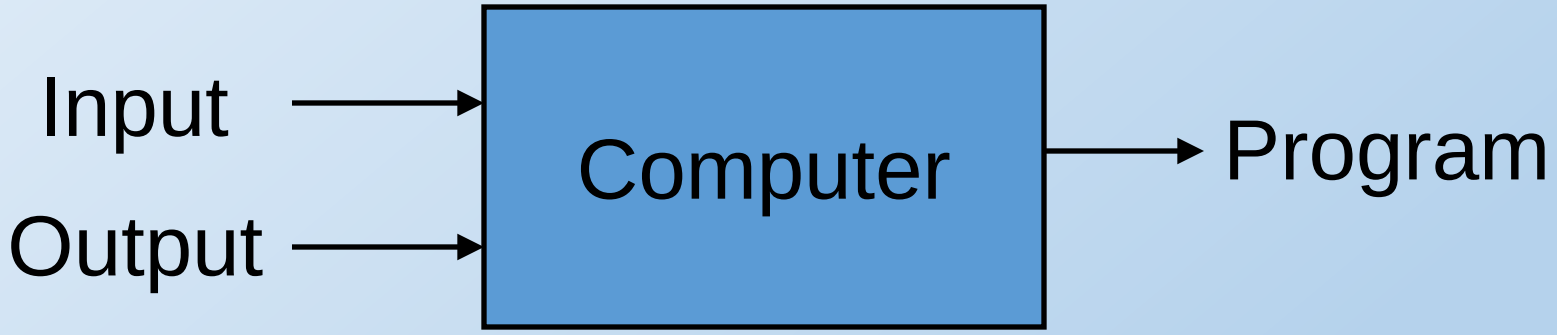
Traditional Programming



Traditional Programming



Machine Learning



Sample Applications

- Web search
- Computational biology
- Finance
- E-commerce
- Space exploration
- Robotics
- Information extraction
- Social networks
- Debugging

ML in a Nutshell

- Tens of thousands of machine learning algorithms
- Hundreds new every year
- Every machine learning algorithm has three components:
  - **Representation**
  - **Evaluation**
  - **Optimization**

Representation

How to represent what is learned in a way that a computer can interpret

- Decision trees
- Sets of rules / Logic programs
- Instances
- Graphical models (Bayes/Markov nets)
- Neural networks
- Support vector machines
- Model ensembles
- Etc.

Evaluation

How to predict how well what was learned will work

- Accuracy
- Precision and recall
- Mean squared error
- Mean absolute error
- Cost / Utility
- Margin
- Entropy
- Etc.

Optimization

How to improve the performance of what is learned

- Combinatorial optimization
  - E.g.: Greedy search
- Convex optimization
  - E.g.: Gradient descent
- Constrained optimization
  - E.g.: Linear programming

Types of Learning

- **Supervised (inductive) learning**
  - Training data includes desired outputs
- **Unsupervised learning**
  - Training data does not include desired outputs
- **Semi-supervised learning**
  - Training data includes a few desired outputs
- **Reinforcement learning**
  - Rewards from sequence of actions

Supervised Learning

- **Given** examples of a function *(X, F(X))*
- **Predict** function *F(X)* for new examples *X*
  - Discrete *F(X)*: Classification
  - Continuous *F(X)*: Regression
  - *F(X)* = Probability(*X*): Probability estimation

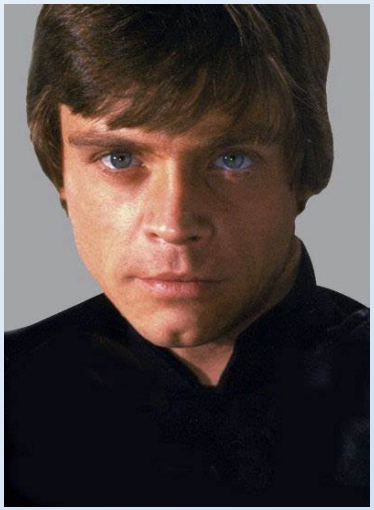
Supervised Learning - Training

x	**f(x)**
• An over-long, dated, old-fashioned war drama, with an overkill of digital effects and tedium.	Negative
• A spectacular combination of true historical story, eye-popping visual effects and massively scaled action sequences, and gripping human drama.	Positive

Supervised Learning - Testing

x	**f(x)**
• An uncommonly awful war picture	??
• It becomes clear almost immediately that *Midway*‘s biggest problem is its emphasis on aggressively one-dimensional characters, as filmmaker Roland Emmerich suffuses the proceedings with one comically stereotypical figure after another	??

Supervised Learning



f(x)

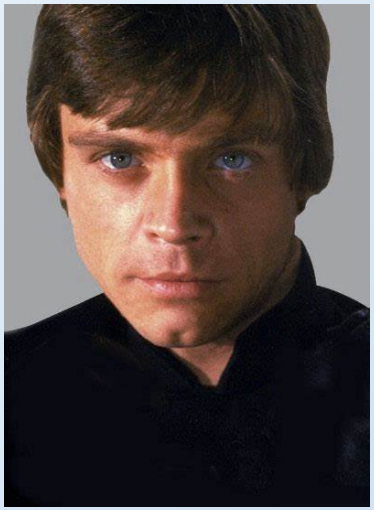
25



f(x)

68

Supervised Learning



f(x)

0 (Male)

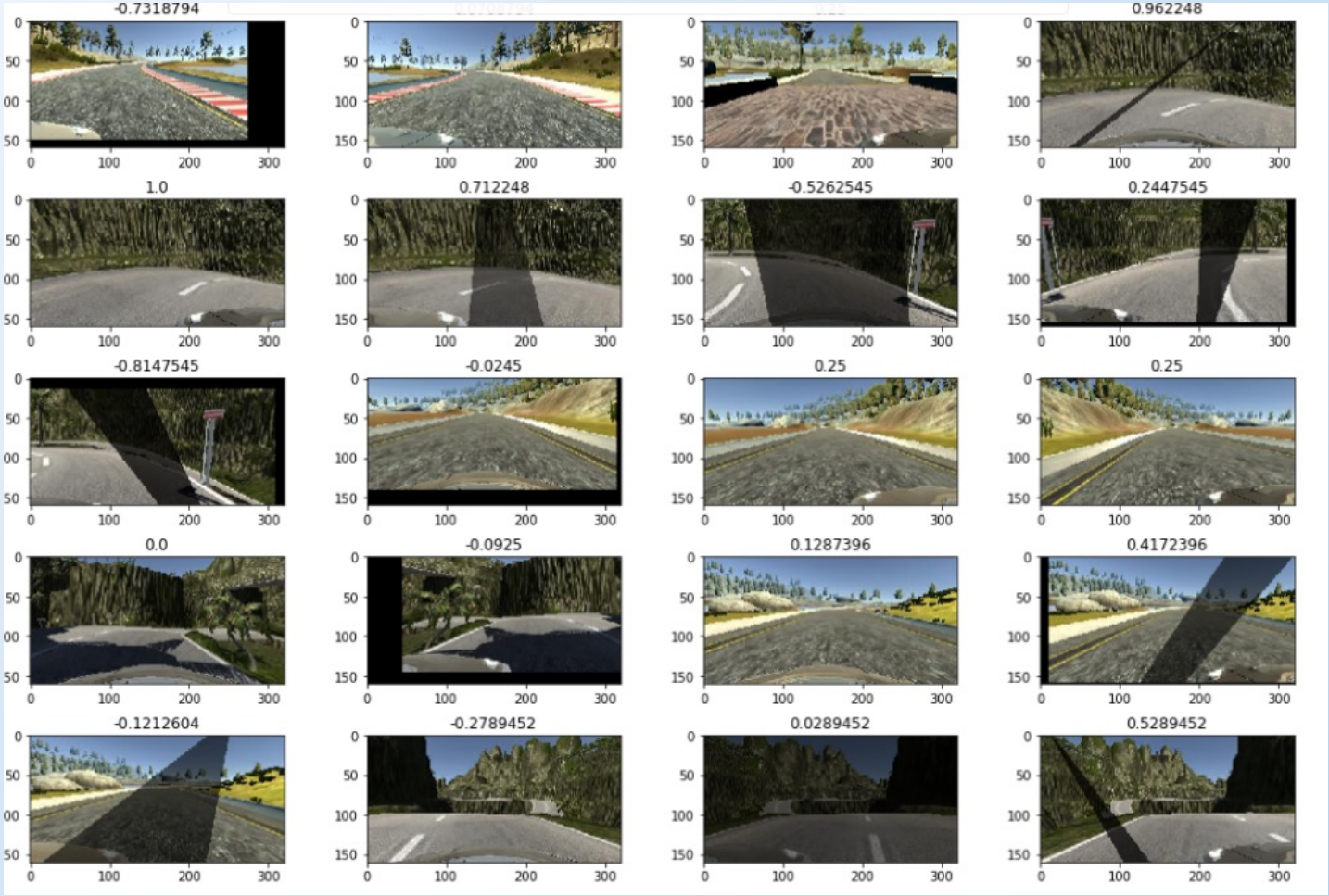


f(x)

1 (female)

Supervised Learning

x



f(x)

Dir = -10

Acc = 0

Dir = 10

Acc = 5

Supervised Learning

x



f(x)

Dir = ?

Acc = ?

Supervised Learning

x

The UTEP Miners [ ]

f(x)

Probability of being the next word in the sentence, for every word in the English language

f('lost') = 0.009

f('won') = 0.001

f('offense') = 0.0005

f('recursion') = 0.0000001

f('platypus') = 0.000000001

...

What We'll Cover

- **Supervised learning**
  - Decision tree induction
  - Instance-based learning
  - Bayesian learning
  - Neural networks
  - Support vector machines
  - Model ensembles
- **Unsupervised learning**
  - Clustering
  - Dimensionality reduction
- **Applications**
  - Natural language processing
  - Computer vision

What We'll Cover

- **We will program several of the algorithms**
- **We will use Python**
- **Some tools: sklearn, tensorflow, pytorch**
- **Course project**

Warning!

This class is programming-intensive

My belief:

“If I can’t program it, I don’t understand it.”

The use of Python allows to program non-trivial algorithms with few lines of code

A Python program is similar to pseudocode; it is critically important to be able to read code, in particular in the GenAI times.

Why program if GenAI can do it for us?

Pair Exercise

For each task, identify *x*, *f(x)* and the type of prediction that needs to be made (classification, regression, or probability estimation)

1. Intelligent cruise control
2. Face tagging
3. Targeted advertising
4. Spam detection
5. Speech recognition
6. Text generation – for example, answering a question or writing a summary using an LLM